2023-2026 AI Report
2023-2026 AI 报告

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一、报告速览 / I. Report Overview

本报告收录 5 篇论文，分为 2 个研究热点（其中高被引 5 篇）。以下按热点逐篇概述其核心内容：
This report includes 5 papers across 2 research hotspots (5 highly cited). Each hotspot and its papers are summarized below:

- 「深度学习」(2 篇 / 2 papers):
  - 《Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence》: Recent years have seen a tremendous growth in Artificial Intelligence (AI)-based methodological development in a broad range of domains.
  - 《New Era of Artificial Intelligence in Education: Towards a Sustainable Multifaceted Revolution》: The recent high performance of ChatGPT on several standardized academic tests has thrust the topic of artificial intelligence (AI) into the mainstream conversation about the future of education.
- 「其他」(3 篇 / 3 papers):
  - 《Revolutionizing healthcare: the role of artificial intelligence in clinical practice》: Healthcare systems are complex and challenging for all stakeholders, but artificial intelligence (AI) has transformed various fields, including healthcare, with…
  - 《Generative artificial intelligence》: Recent developments in the field of artificial intelligence (AI) have enabled new paradigms of machine processing, shifting from data-driven, discriminative AI…
  - 《Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning》: Since its maiden release into the public domain on November 30, 2022, ChatGPT garnered more than one million subscribers within a week. The generative AI tool ChatGPT took the world by surprise with it sophisticated capacity to carry out remarkably complex tasks.

二、分类论文展示 / II. Classified Paper Display

热点一：深度学习 / Hotspot 1: 深度学习

热点主题介绍 / Hotspot intro：本热点聚焦「深度学习」方向，共收录 2 篇论文；代表性工作为《Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence》（2023，引用 1863）。

本方向共收录 / Papers in this direction 2 篇。

1. Interpreting Black-Box Models: a Review on Explainable Artificial Intelligence

- 作者 / Authors: Vikas Hassija, Vinay Chamola, Atmesh Mahapatra 等 / et al.
- 发表时间 / Published: 2023
- 发表期刊 / Venue: Cognitive Computation
- 引用量 / Citations: 1863
- DOI / DOI: `10.1007/s12559-023-10179-8`
- 链接 / Link: `https://openalex.org/W4386142022`

解决的问题 / Problem: 随着机器学习与深度学习模型在各领域的快速扩散，这些固有复杂的模型缺乏对决策过程的解释，被视作“ Black-Box ”。在银行、医疗等关键任务领域，解释其学习与决策过程以实现透明度和可预测性变得极其困难，且寻找模型缺陷以减少假阴性与假阳性结果依然低效，这构成了模型落地的核心瓶颈。 With the rapid proliferation of ML and DL models across domains, these inherently complex models lack explanations for their decision-making processes, being termed as 'Black-Box'. In mission-critical domains like banking and healthcare, explaining their learning and decision processes to achieve transparency and easy predictability has become extremely difficult, and finding flaws to reduce false negative and false positive outcomes remains inefficient, constituting a core bottleneck for model deployment.

现有方案 / Existing approaches: （未明确提及） (not explicitly mentioned)

新方案 / New approach: 本文作为一篇综述，系统性地梳理了解释“ Black-Box ”模型的最新可解释人工智能（XAI）研究进展。1. 对当前最先进的 XAI 研究进行了精心筛选与细致分析，回顾了 XAI 的发展历程。2. 对现有的 XAI 框架及其有效性提供了全面且深入的评估，为应用和理论研究者提供起点。3. 突出了 XAI 研究中新兴的关键问题，展示了针对特定模型以实现更好解释、增强透明度和提高预测准确性的主要趋势。 As a review, this paper systematically collates the latest Explainable Artificial Intelligence (XAI) research progress in interpreting 'Black-Box' models. 1. It meticulously reviews the development of XAI through careful selection and analysis of the current state-of-the-art XAI research. 2. It provides a comprehensive and in-depth evaluation of existing XAI frameworks and their efficacy, serving as a starting point for applied and theoretical researchers. 3. It highlights emerging and critical issues in XAI research, showcasing major, model-specific trends for better explanation, enhanced transparency, and improved prediction accuracy.

效果及局限性 / Results & limitations: （未明确提及） (not explicitly mentioned)

研究价值与应用 / Value & applications: 医疗健康

APA 引用 / APA Citation:

> Vikas Hassija, Vinay Chamola, Atmesh Mahapatra, Abhinandan Singal, Divyansh Goel, Kaizhu Huang, Simone Scardapane, Indro Spinelli, Mufti Mahmud, & Amir Hussain (2023). Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence. Cognitive Computation. https://doi.org/10.1007/s12559-023-10179-8

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2. New Era of Artificial Intelligence in Education: Towards a Sustainable Multifaceted Revolution

- 作者 / Authors: Firuz Kamalov, David Santandreu Calonge, Ikhlaas Gurrib
- 发表时间 / Published: 2023
- 发表期刊 / Venue: Sustainability

- 引用量 / Citations: 967
- DOI / DOI: `10.3390/su151612451`
- 链接 / Link: `https://openalex.org/W4385878593`

解决的问题 / Problem: 随着深度学习技术（如 ChatGPT）在标准化学术测试中展现出高性能，教育系统正面临教学范式转移的紧迫挑战。其核心难点在于，如何在拥抱这项颠覆性技术的同时，确保 AI 驱动技术在学校的可持续发展和部署，并有效防范其潜在滥用风险。 As deep learning technologies (e.g., ChatGPT) demonstrate high performance on standardized academic tests, the education system faces the urgent challenge of a teaching paradigm shift. The essential difficulty lies in how to embrace this disruptive technology while ensuring the sustainable development and deployment of AI-driven technologies in schools, effectively preventing potential abuse.

现有方案 / Existing approaches: （未明确提及） (Not explicitly mentioned)

新方案 / New approach: 本文提出了一种多维度的 AI 教育影响分析框架，具体架构拆解为：1. 应用维度：聚焦人机协同的师生学习、智能辅导系统、自动化评估与个性化学习；2. 优势维度：系统梳理 AI 技术为教育带来的具体增益；3. 挑战维度：全面剖析潜在负面影响与伦理问题，并探索未来实施路径。 The paper proposes a multifaceted analytical framework for AI's impact on education, decomposed as: 1. Application axis: focusing on collaborative teacher-student learning, intelligent tutoring systems, automated assessment, and personalized learning; 2. Advantage axis: systematically outlining specific benefits brought by AI technologies; 3. Challenge axis: comprehensively analyzing potential negative aspects and ethical issues, while exploring future implementation routes.

效果及局限性 / Results & limitations: （未明确提及） (Not explicitly mentioned) 约束：本文为文献综述研究，未提供具体的量化指标、百分比或数据集验证结果，仅得出必须结合 “ 防护措施 ” 来接纳新技术的定性结论。 Limitation: As a literature review, this paper does not provide specific quantitative metrics, percentages, or dataset validation results, only reaching the qualitative conclusion that the new technology must be embraced alongside implemented guardrails.

研究价值与应用 / Value & applications: 通用研究

APA 引用 / APA Citation:

> Firuz Kamalov, David Santandreu Calonge, & Ikhlaas Gurrib (2023). New Era of Artificial Intelligence in Education: Towards a Sustainable Multifaceted Revolution. Sustainability. https://doi.org/10.3390/su151612451

整体分析 / Overall Analysis: 本热点 2 篇论文围绕「深度学习」展开；代表性工作《Interpreting Black-Box Models: A Review 》（2023，引用 1863）贡献突出，整体属新兴/近期热点方向。

奠基性参考论文 / Foundational References:

- Ashish Vaswani 等 (2017): Attention is All you Need —— 被本热点 1 篇引用，全球引用 187789
- H. Arksey, L. O ’ Malley (2005): Scoping studies: towards a methodological framework —— 被本热点 1 篇引用，全球引用 32333
- A. Tricco 等 (2018): PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation —— 被本热点 1 篇引用，全球引用 32267

热点二：其他 / Hotspot 2: 其他

热点主题介绍 / Hotspot intro：本热点聚焦「其他」方向，共收录 3 篇论文；代表性工作为《Revolutionizing healthcare: the role of artificial intelligence in clinical practice》（2023，引用 2938）。

本方向共收录 / Papers in this direction 3 篇。

1. Revolutionizing Healthcare: the Role of Artificial Intelligence in Clinical Practice

- 作者 / Authors: Shuroug A. Alowais, Sahar S. Alghamdi, Nada Alsuhebany 等 / et al.
- 发表时间 / Published: 2023
- 发表期刊 / Venue: BMC Medical Education
- 引用量 / Citations: 2938
- DOI / DOI: `10.1186/s12909-023-04698-z`
- 链接 / Link: `https://openalex.org/W4386958277`

解决的问题 / Problem: 医疗系统的高度复杂性给所有利益相关者带来了严峻挑战，其本质难点在于如何将人工智能有效且负责任地整合到临床实践中，以真正提升患者护理质量与生活水平，同时必须妥善应对数据隐私、算法偏见以及对人类专业知识依赖等核心痛点。 Healthcare systems' high complexity poses severe challenges to all stakeholders, with the essential difficulty lying in how to effectively and responsibly integrate AI into clinical practice to genuinely improve patient care and quality of life, while properly addressing core pain points like data privacy, algorithmic bias, and the reliance on human expertise.

现有方案 / Existing approaches: （未明确提及） (Not explicitly mentioned)

新方案 / New approach: 本文作为综述性研究，通过系统梳理 PubMed/Medline、Scopus 和 EMBASE 等索引文献，构建了以下多维度的 AI 临床应用架构：1. 疾病诊断与实验室测试优化，利用大型数据集识别模式以超越人类表现；2. 个性化治疗与用药剂量优化，辅助临床决策并制定个性化治疗计划；3. 患者参与度提升，涵盖虚拟健康助手、心理健康支持、患者教育及对医患信任的影响评估。 As a review study, this paper constructs the following multi-dimensional architecture for AI clinical applications by systematically reviewing indexed literature such as PubMed/Medline, Scopus, and EMBASE: 1. Disease diagnosis and clinical laboratory testing optimization, leveraging large datasets to identify patterns to surpass human performance; 2. Personalized treatment and medication dosage optimization, assisting clinical decision-making and developing personalized treatment plans; 3. Patient engagement enhancement, covering virtual health assistants, mental health support, patient education, and the impact on patient-physician trust.

效果及局限性 / Results & limitations: 研究表明，AI 在提升疾病诊断准确性、降低医疗成本、节省时间以及减少人为错误方面展现出巨大潜力，能够有效支持个性化医疗、人群健康管理及临床指南制定。（未明确提及具体的量化指标、百分比、数据集规模或实际部署证据）。约束：AI 在医疗中的负责任与有效实施仍受限于数据隐私问题、算法偏见以及不可替代的人类专业知识需求。 Research indicates that AI shows great potential in improving the accuracy of disease diagnosis, reducing medical costs, saving time, and minimizing human errors, effectively supporting personalized medicine, population health management, and clinical guideline development. (Specific quantitative metrics, percentages, dataset sizes, or actual deployment evidence were not explicitly mentioned). Limitation: The responsible and effective implementation of AI in healthcare is still constrained by data privacy issues, algorithmic bias, and the indispensable need for human expertise.

研究价值与应用 / Value & applications: 医疗健康、推荐系统、科学发现

APA 引用 / APA Citation:

> Shuroug A. Alowais, Sahar S. Alghamdi, Nada Alsuhebany, Tariq Alqahtani, Abdulrahman Alshaya, Sumaya N. Almohareb, Atheer Aldairem, Mohammed Alrashed, Khalid Bin Saleh, Hisham A. Badreldin, Majed S. Al Yami, Shmeylan Al Harbi, & Abdulkareem Albekairy (2023). Revolutionizing healthcare: the role of artificial intelligence in clinical practice. BMC Medical Education. https://doi.org/10.1186/s12909-023-04698-z

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2. Generative Artificial Intelligence

- 作者 / Authors: Leonardo Banh, Gero Strobel
- 发表时间 / Published: 2023
- 发表期刊 / Venue: Electronic Markets
- 引用量 / Citations: 502
- DOI / DOI: `10.1007/s12525-023-00680-1`
- 链接 / Link: `https://openalex.org/W4389359039`

解决的问题 / Problem: 人工智能领域正面临从传统的数据驱动、判别式任务向复杂创造性任务转型的核心挑战，其本质难点在于如何让机器基于简单的用户提示，跨领域生成既新颖又逼真的内容，同时要求研究者和从业者深刻理解其独特性质以平衡潜力与风险。 Artificial intelligence faces a core challenge of transitioning from traditional data-driven, discriminative tasks to complex creative tasks, with the essential difficulty lying in enabling machines to generate novel and realistic content across domains based on basic user prompts, while requiring researchers and practitioners to deeply understand its unique nature to balance potential and risks.

现有方案 / Existing approaches: （未明确提及） (Not explicitly mentioned)

新方案 / New approach: 本文作为一篇全面的综述，其创新内容在于系统性地拆解了生成式人工智能的基础架构：1. 概念性引入相关术语与技术；2. 概述构成生成式 AI 的固有属性；3. 详细阐述其未来的潜力与面临的挑战。 As a comprehensive overview, the paper's novelty lies in systematically deconstructing the foundational architecture of generative AI: 1. conceptually introducing relevant terms and techniques; 2. outlining the inherent properties that constitute generative AI; 3. elaborating on its future potentials and challenges.

效果及局限性 / Results & limitations: （未明确提及） (Not explicitly mentioned)

研究价值与应用 / Value & applications: 自然语言处理、计算机视觉

APA 引用 / APA Citation:

> Leonardo Banh, & Gero Strobel (2023). Generative artificial intelligence. Electronic Markets. https://doi.org/10.1007/s12525-023-00680-1

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3. Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning

- 作者 / Authors: David Baidoo-Anu, Leticia Owusu Ansah
- 发表时间 / Published: 2023
- 发表期刊 / Venue: Journal of AI
- 引用量 / Citations: 1676

- DOI / DOI: `10.61969/jai.1337500`
- 链接 / Link: `https://openalex.org/W4389437528`

解决的问题 / Problem: 生成式人工智能（如 ChatGPT）在执行复杂教育任务时展现出的非凡能力，引发了教育工作者对现有教育实践被彻底颠覆的复杂情绪与核心挑战。其本质难点在于，如何在拥抱这项可能革命性改变教学与学习的技术的同时，妥善应对其带来的不确定性。此外，如何安全且富有建设性地将这些不断演进的生成式 AI 工具整合到教育生态中，也是亟待解决的痛点。 Generative AI like ChatGPT exhibits extraordinary abilities in performing complex educational tasks, triggering mixed feelings among educators regarding the potential disruption of existing educational praxis. The essential difficulty lies in how to embrace this technology—which may revolutionize teaching and learning—while properly managing the associated uncertainties. Furthermore, safely and constructively integrating these evolving generative AI tools into the educational ecosystem remains a critical pain point.

现有方案 / Existing approaches: （未明确提及）(Not explicitly mentioned)

新方案 / New approach: 本文作为一项探索性研究，通过综合近期的现有文献，系统梳理了 ChatGPT 在促进教学与学习方面的潜在益处与缺陷。具体分析架构包括：1. 识别促进个性化和交互式学习的潜力；2. 探讨其为形成性评估活动生成提示词以提供持续反馈的功能；3. 明确指出其固有缺陷，如生成错误信息、训练数据偏见加剧现有偏见以及隐私问题等。 As an exploratory study, this paper synthesizes recent extant literature to systematically outline the potential benefits and drawbacks of ChatGPT in promoting teaching and learning. The specific analytical framework includes: 1. identifying the potential for promoting personalized and interactive learning; 2. exploring its function in generating prompts for formative assessment activities that provide ongoing feedback; 3. explicitly pointing out inherent drawbacks such as generating wrong information, training data biases that may augment existing biases, and privacy issues.

效果及局限性 / Results & limitations: 本文指出 ChatGPT 自 2022 年 11 月 30 日发布后一周内即获得超过一百万订阅用户，以此作为其广泛影响力的量化证据。约束：该研究为基于文献的探索性综述，未提供具体的教学实验指标或实际部署数据来验证其提出的潜在益处。 The paper notes that ChatGPT garnered more than one million subscribers within a week of its release on November 30, 2022, serving as quantitative evidence of its broad impact. Limitation: This study is a literature-based exploratory survey and does not provide specific teaching experiment metrics or actual deployment data to validate the proposed potential benefits.

研究价值与应用 / Value & applications: 推荐系统

APA 引用 / APA Citation:

> David Baidoo-Anu, & Leticia Owusu Ansah (2023). Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning. Journal of AI. https://doi.org/10.61969/jai.1337500

整体分析 / Overall Analysis: 本热点 3 篇论文围绕「其他」展开；代表性工作《Revolutionizing healthcare: the role of 》（2023，引用 2938）贡献突出，整体属新兴/近期热点方向。

奠基性参考论文 / Foundational References:

- A. Esteva 等 (2017): Dermatologist – level classification of skin cancer with deep neural networks —— 被本热点 1 篇引用，全球引用 12941
- Michael I. Jordan, T. Mitchell (2015): Machine learning: Trends, perspectives, and prospects —— 被本热点 1 篇引用，全球引用 8515

- E. Topol (2019): High-performance medicine: the convergence of human and artificial intelligence —— 被本热点 1 篇引用，全球引用 8027

三、研究趋势 / III. Research Trends

未来研究趋势 / Future Trends

- 本批论文覆盖多个相关方向，整体研究活跃 / These papers span several related directions with active research

可深挖方向 / Directions Worth Exploring

- 生成式 AI（如 ChatGPT）在教育场景中的可解释性研究：将 XAI 技术引入教学辅助系统，使 AI 的生成逻辑和反馈机制对师生透明，以解决黑盒模型在严谨教育环境中的信任与有效性问题。 / Explainability of Generative AI (e.g., ChatGPT) in educational scenarios: Integrating XAI techniques into teaching-assistant systems to make the generative logic and feedback mechanisms transparent to teachers and students, addressing the trust and effectiveness issues of black-box models in rigorous educational environments.
- 生成式 AI 在临床医疗与教育领域的跨领域可迁移性评估：对比分析生成式 AI 在医疗诊断与个性化教学中的共性应用模式（如个性化生成、交互式辅助）， 提炼出一套可跨领域复用的生成式 AI 落地评估框架。 / Cross-domain transferability assessment of Generative AI in clinical healthcare and education: Comparatively analyzing the common application patterns of Generative AI in medical diagnosis and personalized teaching (e.g., personalized generation, interactive assistance) to extract a reusable implementation evaluation framework across domains.

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